

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2022
Department of Materials Science and Engineering
MSE 3101
Transport Phenomena in Materials

Time: 3 hours

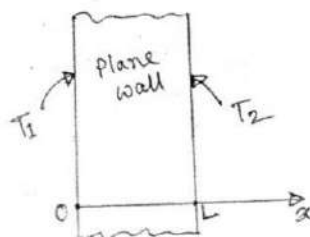
Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Derive the Bernoulli equation for steady flow along a streamline in the inviscid regions of flow with a suitable figure. (15)
(b) Through a refinery, fuel ethanol is flowing in a pipe at a velocity of 2 m/s and a pressure of 102,500 Pa. The refinery needs the ethanol to be at a pressure of 203,800 Pa on a lower level. How far must the pipe drop in height in order to achieve this pressure? Assume that the velocity does not change. (The density of ethanol is 790 kg/m³). (10)
(c) The viscosity of a fluid is to be measured by a viscometer constructed of two 40 cm long concentric cylinders. The outer diameter of the inner cylinder is 14 cm, and the gap between the two cylinders is 0.13 cm. The inner cylinder is rotated at 320 rpm, and the torque is measured to be 1.9 N.m. Determine the viscosity of the fluid. (10)
2. (a) Starting with an energy balance on a cylindrical shell volume element, derive the steady one-dimensional heat conduction equation for a long cylinder with constant thermal conductivity in which heat is generated at a rate of e_{gen} . (17)
(b) Consider a large plane wall of thickness $L = 0.23$ m, thermal conductivity $k = 1.3$ W/m.°C, and surface area $A = 15$ m². The two sides of the wall are maintained at constant temperatures of $T_1 = 125^\circ\text{C}$ and $T_2 = 50^\circ\text{C}$, respectively, as shown in Figure 2(b). Determine (a) the variation of temperature within the wall and the value of temperature at $x = 0.1$ m and (b) the rate of heat conduction through the wall under steady conditions. (18)



3. (a) A hollow sphere of inside radius $r = a$ and outside radius $r = b$ is electrically heated at the inner surface at a constant rate of q_o W/m². At the outer surface it dissipates heat by convection into a fluid at temperature T_∞ with a heat transfer coefficient h . The thermal conductivity k of the solid is constant. Develop expressions for the determination of the inner and outer surface temperatures T_1 and T_2 of the sphere. Calculate the inner and outer surface temperatures for $a = 3$ cm, $b = 5$ cm, $h = 400$ W/m².°C, $T_\infty = 100^\circ\text{C}$, $k = 15$ W/m.°C, and $q_o = 10^5$ W/m². (18)

- (b) Steam at $T_{\alpha 1} = 320^\circ\text{C}$ flows in a cast iron pipe ($k = 80\text{ W/m}\cdot\text{K}$) whose inner and outer diameters are $D_1 = 5\text{ cm}$ and $D_2 = 5.5\text{ cm}$, respectively. The pipe is covered with 3 cm thick glass wool insulation with $k = 0.05\text{ W/m}\cdot\text{K}$. Heat is lost to the surroundings at $T_{\alpha 2} = 5^\circ\text{C}$ by natural convection and radiation, with a combined heat transfer coefficient of $h_2 = 18\text{ W/m}^2\cdot\text{K}$. Taking the heat transfer coefficient inside the pipe to be $h_1 = 60\text{ W/m}^2\cdot\text{K}$, determine the rate of heat loss from the steam per unit length of the pipe. Also, determine the temperature drops across the pipe shell and the insulation. (17)

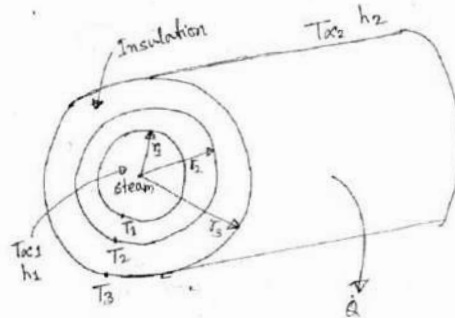


Figure: 3(b)

4. (a) The inner surface at $r = a$ and outer surface at $r = b$ of a hollow cylinder are maintained at uniform temperature T_1 and T_2 respectively. (i) Develop an expression for the one dimensional steady state temperature distribution $T(r)$ in the cylinder. (ii) Develop an expression for the thermal resistance of a hollow cylinder of length H . (18)
- (b) A long 20 cm diameter cylindrical shaft made of stainless steel 304 comes out of an oven at a uniform temperature of 600°C . The shaft is then allowed to cool slowly in an environment chamber at 200°C with an average heat transfer coefficient of $h = 80\text{ W/m}^2\cdot\text{K}$. The properties of stainless steel 304 are $k = 14.9\text{ W/m}\cdot\text{K}$, $\rho = 7900\text{ kg/m}^3$, $C_p = 477\text{ J/kg}\cdot\text{K}$, and $\alpha = 3.95 \times 10^{-6}\text{ m}^2/\text{s}$. Determine the temperature at the center of the shaft 45 min after the start of the cooling process. Also determine the heat transfer per unit length of the shaft during this time period. [Use the charts in Figure 4(b) if necessary]. (17)

Section B

(Answer ANY THREE questions from this section in answer script B)

5. (a) Determine an expression for the net radiative heat flux at the surface of the small solid object of following Figure 5(a) in terms of the surface and surroundings temperature and the Stefan-Boltzmann constant. The small object is a blackbody. (15)
- (b) Consider a large isothermal enclosure that is maintained at a uniform temperature of 2000 K . (10)
- (i) Calculate the emissive power of the radiation that emerges from a small aperture on the enclosure surface.
- (ii) What is the wavelength λ below which 20% of the emission is concentrated?
- (iii) What is the maximum emissive power and wavelength at which this emission occurs? [Use Table 5(b)].
- (c) The spectral distribution of surface irradiation is shown in Figure 5(c). What is the total irradiation? (10)

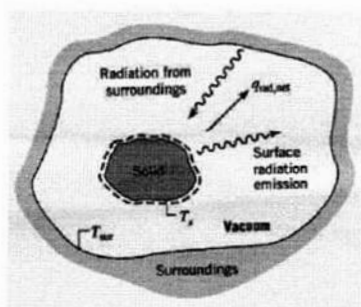


Figure: 5(a)

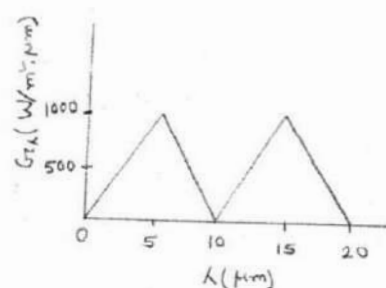
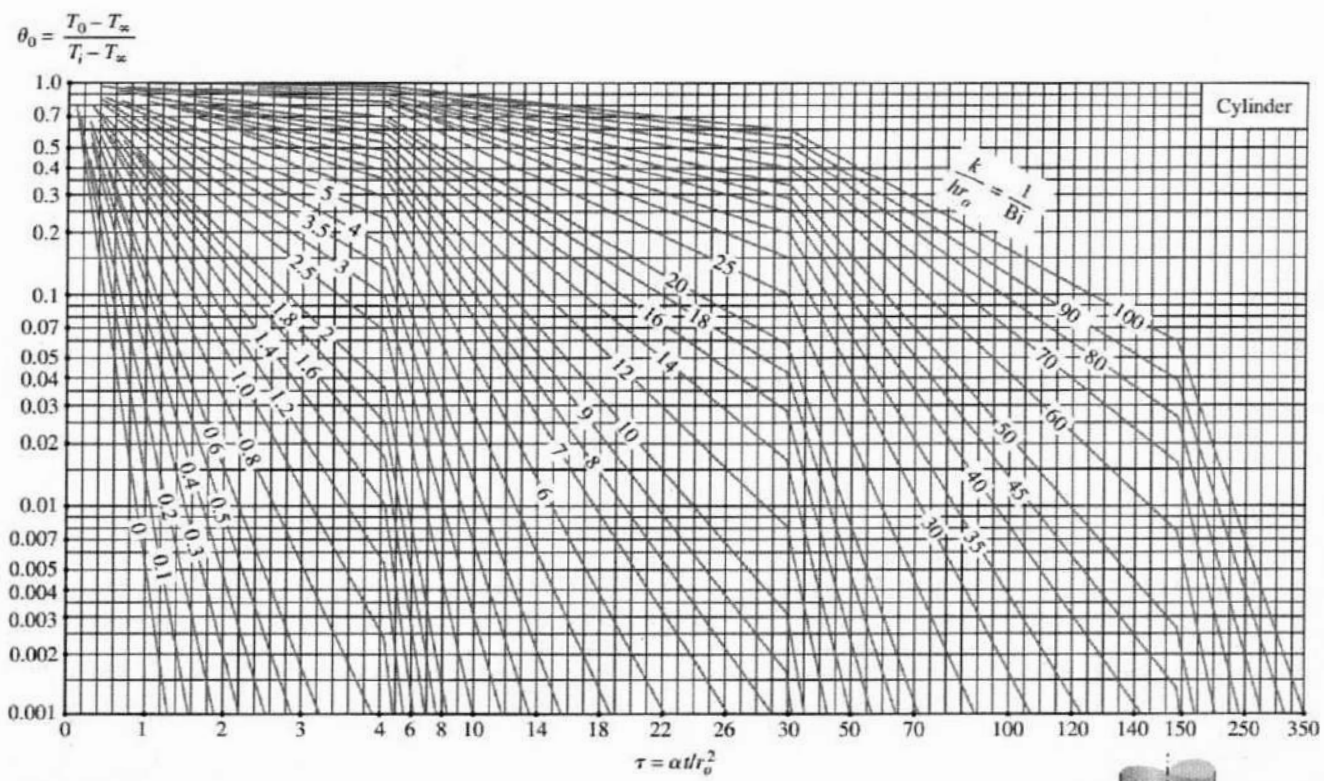


Figure: 5(c)

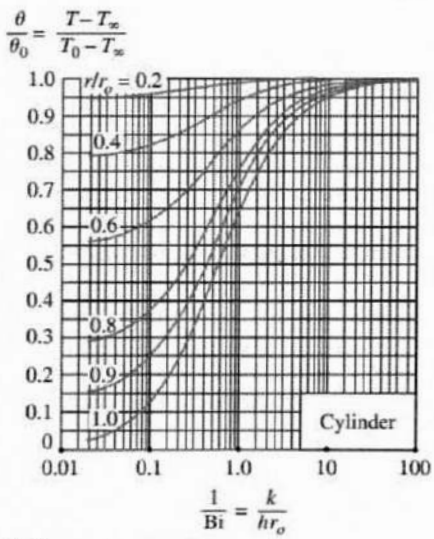
6. (a) Derive an expression for the log mean temperature difference (LMTD) for a counter flow heat exchanger. (15)
- (b) Prove that counter flow heat exchanger is better than parallel heat exchanger. Use the values $T_{h,i} = 280^\circ\text{C}$, $T_{h,o} = 140^\circ\text{C}$, $T_{c,i} = 70^\circ\text{C}$, $T_{c,o} = 115^\circ\text{C}$. (15)
- (c) Draw a temperature distribution for counter heat flow exchanger having equivalent fluid heat capacities, $C_h = C_c$. (05)

7. (a) Consider a diffuse circular disk of diameter D and area A_i and a plane diffuse surface of area A_j ($A_i \ll A_j$). The surfaces are parallel and A_i is located at a distance L from the center of A_j . Obtain an expression for the view factor F_{ij} . (12)
- (b) A cryogenic fluid flows through a long tube of 20 mm diameter, the outer surface of which is diffuse and gray with $\varepsilon_1 = 0.03$ and $T_1 = 80\text{K}$. This tube is concentric with a larger tube of 50 mm diameter, the inner surface of which is diffuse and gray with $\varepsilon_2 = 0.06$ and $T_2 = 300\text{K}$. The space between the surfaces is evacuated. Calculate the heat gain by the cryogenic fluid per unit length of tubes. (08)
- (c) Prove that "Radiation exchange between large parallel planes will be double than between the large parallel plane with a radiation shield". (15)

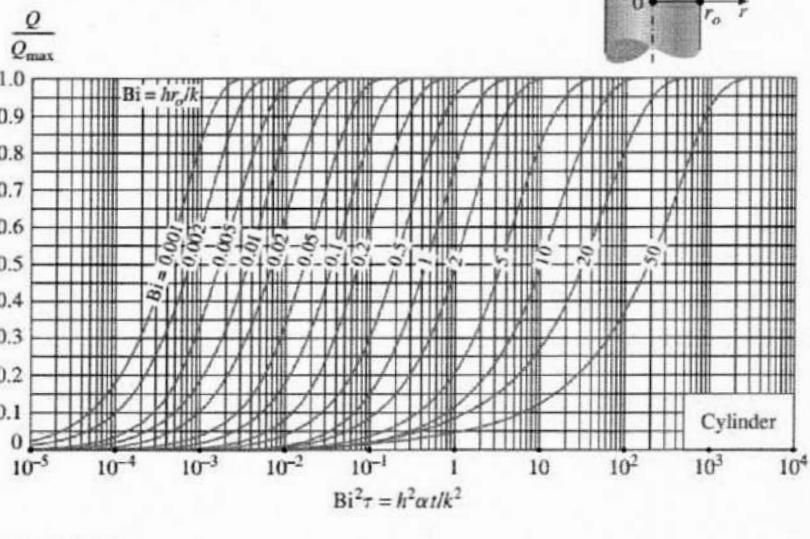
8. (a) Prove that the solid angle associated with the entire hemisphere is 2π sr. (15)
- (b) Consider the diffusion of hydrogen in air at $T = 293\text{K}$. Calculate the species flux on both molar and mass bases, the concentration gradient $dc_A/dx = 1\text{Kmol}/(\text{m}^3.\text{m})$. Here, $D_{AB} = 0.41 \times 10^{-4}\text{m}^2/\text{s}$ at 298K . (10)
- (c) Derive Kirchhoff's law of radiation. (10)



(a) Centerline temperature.
 From M. P. Heisler, "Temperature Charts for Induction and Constant Temperature Heating,"
 Trans. ASME 69, 1947, pp. 227-36. Reprinted by permission of ASME International.



(b) Temperature distribution.



(c) Heat transfer.

Figure 4(a) Transient temperature and heat transfer charts for a cylinder of radius r_o initially at a uniform temperature T_i subjected to convection from all sides to an environment at temperature T_∞ with a convection coefficient of h .

λT ($\mu\text{m} \cdot \text{K}$)	$F_{(0 \rightarrow \lambda)}$	$I_{\lambda,b}(\lambda, T) / \sigma T^5$ ($\mu\text{m} \cdot \text{K} \cdot \text{sr})^{-1}$	$\frac{I_{\lambda,b}(\lambda, T)}{I_{\lambda,b}(\lambda_{\text{max}}, T)}$
200	0.000000	0.375034×10^{-27}	0.000000
400	0.000000	0.490335×10^{-13}	0.000000
600	0.000000	0.104046×10^{-8}	0.000014
800	0.000016	0.991126×10^{-7}	0.001372
1,000	0.000321	0.118505×10^{-5}	0.016406
1,200	0.002134	0.523927×10^{-5}	0.072534
1,400	0.007790	0.134411×10^{-4}	0.186082
1,600	0.019718	0.249130	0.344904
1,800	0.039341	0.375568	0.519949
2,000	0.066728	0.493432	0.683123
2,200	0.100888	0.589649×10^{-4}	0.816329
2,400	0.140256	0.658866	0.912155
2,600	0.183120	0.701292	0.970891
2,800	0.227897	0.720239	0.997123
2,898	0.250108	0.722318×10^{-4}	1.000000

Table 5(b)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 3103
(Materials Manufacturing Processes)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Why is meant by expandable mould and permanent mould? Which one will you prefer to produce more complex shape casting products? Justify your answer. (12)
(b) Write down the basic steps involved in making a casting. (10)
(c) Briefly explain the effect of clay in the moulding sand. (08)
(d) What are the general characteristics of greensand moulding? (05)
2. (a) Why does the liquid not transform to solid immediately below equilibrium melting temperature? Explain in brief. (10)
(b) What is meant by 'undercooling'? Examine how the depression of undercooling influence the solidified structure of impure metals and alloy. (10)
(c) Explain the mechanism of equiaxed crystals formation with necessary figure. (10)
(d) Determine the DAS from the given micrograph with magnification 150. See indicated points a, b, c in Figure 2(d). (05)

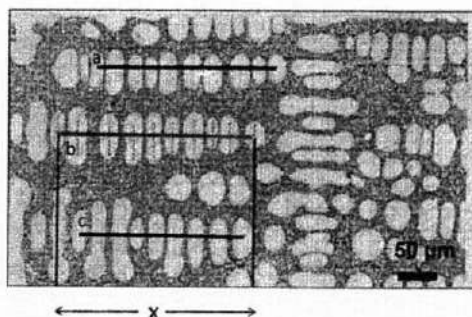
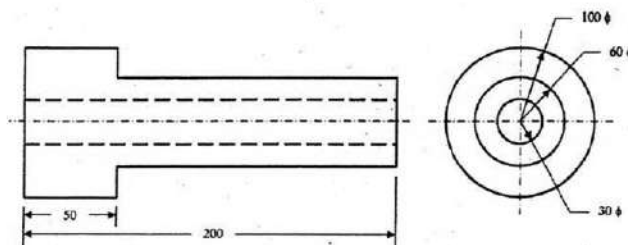


Figure 2(d)

3. (a) State and explain the 'Heat transfer requirement' and 'Mass transfer requirement' of feeding rules. (08)
(b) How can you increase the efficiency of a feeder? (09)
(c) A foundry receives an order of manufacturing a prototype casting shown in Figure 3(c) to be cast using gunmetal. As a foundry engineer of the company,
(i) explain the types of pattern and moulding systems you would use for this job
(ii) using necessary allowances determine the size of the pattern and its core prints
(iii) if the moulding flask allows only a total length of 30mm of core prints, examine the possibility of requiring a chaplet and its area, if required. [See data sheet for Question 3(c)]. (18)



All dimensions are in mm.

Figure 3(c)

4. (a) Compare between pressurized and unpressurized gating system. Select a suitable one and justify your answer. (07)
- (b) For a standard pouring basin ($h=50$ mm) and a sprue height of 150 mm, what is the liquid metal velocity at the base of the sprue? (06)
- (c) What is the objective of pouring basin? Write down the bad effects of using conical cup and the solution of it. (08)
- (d) Design a suitable gating system for a cylindrical pipe (150 mm outer diameter, 15 mm thickness, 200 mm length) made using grey iron of composition 3.0 %C, 1.8 % Si, 0.3% P and density 7.0 g/cc. The moulding system to be used is green sand and pouring temperature is 2450 F. Assume reasonable value for any missing data. List all basis and assumptions you made during the design. Empirical formulae for pouring time (in seconds) calculations of thin walled grey iron casting of thickness 8-15 mm and weight less than 450 kg is $t = 2.2\sqrt{w}$, where, w is the weight of the liquid metal poured. [Use Fluidity chart for Question 4(d)]. (14)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is meant by the pseudo-plastic behavior of polymer melt? Explain in brief. (08)
- (b) Why are thermosetting polymers more harder and stronger than thermoplastic polymers? Explain in brief. (07)
- (c) Which blow molding process will you prefer for producing a strong product? Justify your answer. (13)
- (d) Differentiate between compression molding and transfer molding process. (07)
6. (a) Briefly describe the formation of continuous chip and discontinues chip. (10)
- (b) What is meant by Orthogonal Cutting Model? Draw a force diagram for Orthogonal Cutting Model. (13)
- (c) Write down the importance of powder metallurgy in the production of metallic parts. (12)
7. (a) What is the abbreviation of SLS? Which AM technique will you prefer for ceramic products fabrication? Justify your answer. (13)
- (b) What is meant by hydro plasticity? How do clay minerals show hydro plasticity? Explain in brief. (12)
- (c) Why are drying and firing operations required in ceramic manufacturing process? (10)
8. (a) What are the general properties of ceramics? Classify the body materials of ceramic and their function. (07)
- (b) Compare between thermoplastic polymer and thermosetting polymer. (06)
- (c) With necessary chemical reactions and diagrams, briefly discuss how Portland cement is produced. (12)
- (d) As a materials engineer, select a suitable one for polymer double-layer bottles making between extrusion blow molding and injection blow molding. Justify your answer. (10)

Given data:

Gunmetal: Density = 8.7 g/cm^3 , Volume expansion coefficient = 7.0%

Pouring time, $t = 1.8 (\delta W)^{1/3}$ sec. (δ = average thickness in mm, W = weight of casting in kg)

Coresand: Density = 1.6 g/cm^3 , Compressive strength = 0.35 kg/cm^2 , Factory of safety = 4.

Data for contraction allowances:

Contraction rule	Materials used and place of use
+8/1000	Cast iron in general, part of thin cast iron
+9/1000	Cast iron products of high shrinkage, part of thin cast steel
+10/1000	Same as above, and aluminium
+12/1000	Aluminium alloys, bronze, cast steel (thickness 5-7 mm)
+14/1000	High tension brass, cast steel
+16/1000	Cast steel (thickness over 10 mm in general)
+20/1000	Large cast steel
+25/1000	Large cast iron

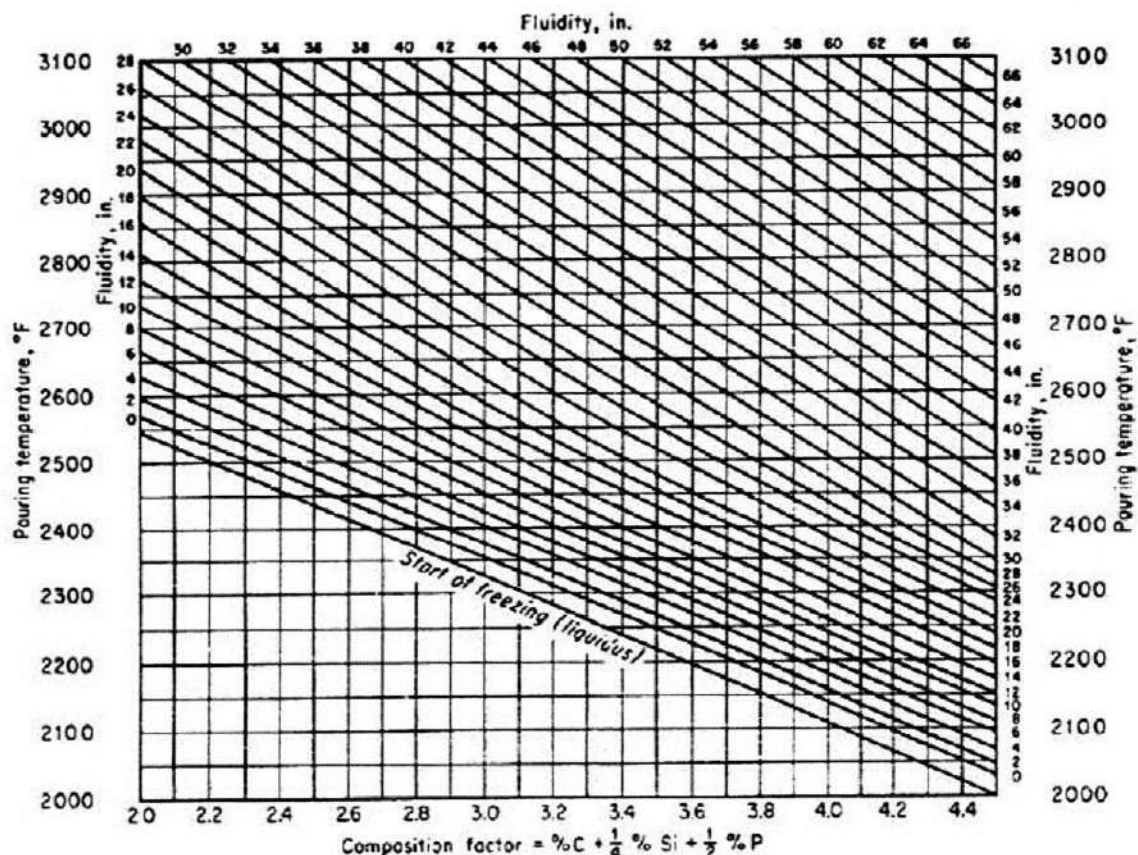
Data for general machining allowances:

Type of metal and alloys	Machining allowance (mm)
Cast irons	
(i) Large size castings (>1000 mm)	10.0
(ii) Medium size castings (<150 mm)	3.0
Cast steels	
(i) Large size castings (>1000 mm)	12.0
(ii) Medium size castings (<150 mm)	4.3
Non-ferrous materials	
(i) Large size castings (>1000 mm)	5.0
(ii) Medium size castings (<150 mm)	1.5

Data for approximate taper allowances:

Height of pattern mm	Shell Moulding	Sand moulding		
		Metal	Wood	
		Machine drawn	Manual drawn	Machine drawn
Up to 20	0° 45'	1° 30'	3°	3°
20 to 50	0° 30'	1°	1° 30'	1° 30'
:	:	:	:	:
100 to 200	0° 20'	0° 30'	0° 45'	0° 45'

Question 3(c): Data Sheet



Question 4(d): Fluidity chart

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 3105
(Welding and Materials Joining Processes)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Explain the importance of learning different materials joining processes. Also list out the parameters that influence the choice of particular joining processes. (10)
 - (b) In which conditions do you choose solid-state diffusion welding instead of the fusion welding process? (10)
 - (c) Describe the conditions of performing Furnace Brazing and explain the working mechanism of Induction-Brazing. (15)
-
2. (a) From the weld symbols shown in figure 2(a), draw the diagram of desired welds. (10)

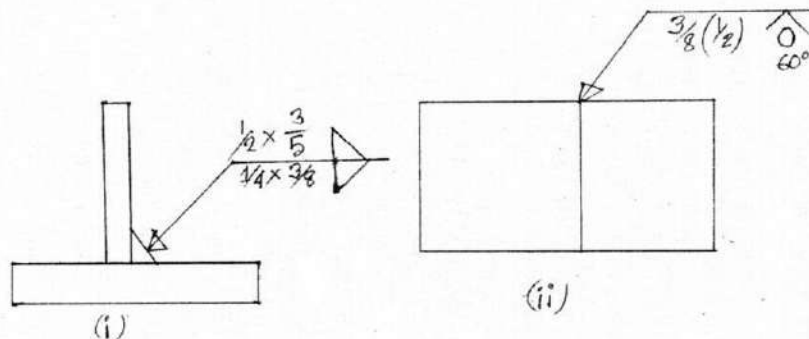


Figure 2(a)

- (b) Discuss the modes of metal transfer in GMAW with necessary figure. Which mode of metal transfer is suitable for out of position welding and why? (15)
 - (c) Compare electroslag welding process with submerged-arc welding process. (10)
-
3. (a) Write down the functions of electrode covering in shielded metal arc welding process. (12)
 - (b) Briefly describe the process of resistance welding. (09)
 - (c) With neat sketches, explain the joint formation in explosion welding. (14)
-
4. (a) Suppose you are assigned in a renowned company as an expert welder to join many long pipes. Which types of friction welding process will you choose and why? Explain that welding process with neat sketches. (12)
 - (b) Explain the process of electron beam welding. Also discuss the advantages of electron beam welding over the conventional welding processes. (13)
 - (c) How can you reduce reflectivity in laser beam welding? (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) What is meant by destructive testing and non-destructive testing of weldments? Explain one of the destructive testing methods for testing weldments. (13)
- (b) Explain stress corrosion cracking in austenitic stainless steel with remedies in HAZ. (10)
- (c) Draw and explain stress and temperature distribution curves along A-A, B-B and C-C section of weldments given in figure 5(c). (12)

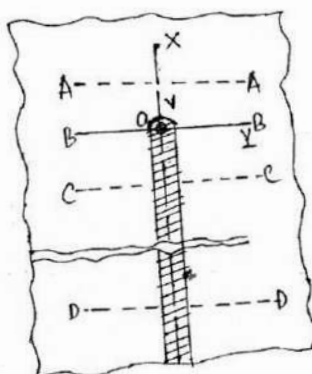


Figure 5(c)

6. (a) Briefly describe the surface nucleation and dendrite fragmentation nucleation mechanism of weld metal. (11)
- (b) Derive the relationship between growth rate and welding speed at the centerline and fusion line. (12)
- (c) Explain the formation of porosity in weld-metal and how to overcome this kind of defect? (12)
7. (a) Explain the liquation mechanism of the C_2 alloy given in figure 7(a) which consists of α matrix and A_xB_y particles in the as-cast condition. (13)

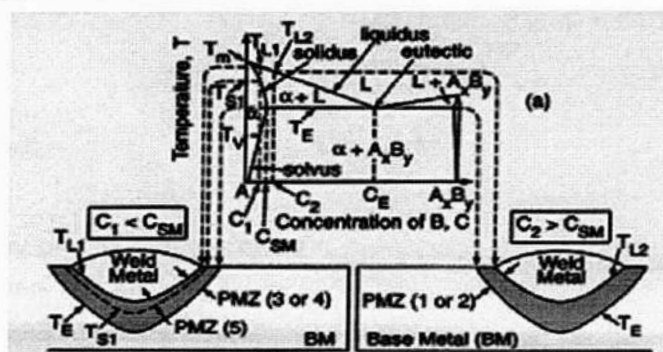


Figure 7(a)

- (b) In which condition will you select the rivet joining process to join materials? Write down the procedure of rivet joining process. (12)
- (c) What is lamellar tearing in a weldment? How does it form? How can lamellar tearing be avoided by modifying the design of weldments? (10)
8. (a) What is the effect of S and P impurities in solidification cracking? How can you reduce this problem? (12)
- (b) Suppose you are assigned in a project as a NDT expert to inspect surface and subsurface defect in joints created by fusion welding. Which NDT technique will you choose and why? Explain the working principle of that technique. (13)
- (c) Write the mechanisms of Diffusion and Glaze bonding in case of ceramics joining processes. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

MSE 3107
(Physical Metallurgy of Materials)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define residual elements of steel. Mention at least four elements that are totally detrimental (10) for the properties of steels and discuss their effects.
- (b) You work in a manufacturing company that specializes in producing thin wire and sheet (10) products. Recently you have been assigned a project to develop a new fabrication procedure for a high strength metal alloy. The procedure involves extensive cold working to achieve the desired shape and dimensions. Considering the specific requirements of your project, choose a suitable heat treatment technique to restore ductility and prevent fracture in the production process and give proper reasoning(s) to your selection.
- (c) Design the annealing and normalizing schedules for 4340 steel bars having 40 mm diameter. (15) Also, calculate the UTS, YS and % elongation of the annealed and normalized steels using figure 1(c).

The chemical composition of 4340 grade steel bars is given below:

AISI-SAE Number	% C	% Mn	% Si	% Ni	% Cr	% Mo
4340	0.38-0.43	0.60-0.80	0.15-0.30	1.65-2.00	0.70-0.90	0.20-0.30

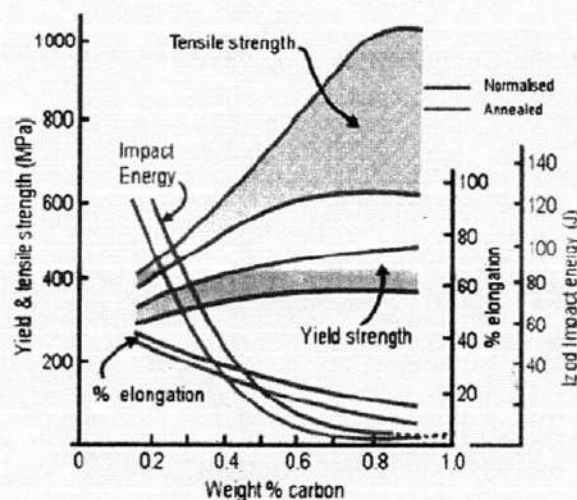


Figure 1(c)

2. (a) Explain the metallurgical aspects of austenitizing with neat sketches. (13)
- (b) Compare between Austempering and Martempering process. (10)

- (c) AISI 1050 steel has the following chemical compositions (in wt %): (12)

$$C=0.45-0.55, Mn=0.60-0.90$$

Now, AISI 1050 steel is to be hardened isothermally to achieve fully upper bainitic microstructures. Design the heat treatment schedule. The transformation diagram of the steel is given in figure 2(c).

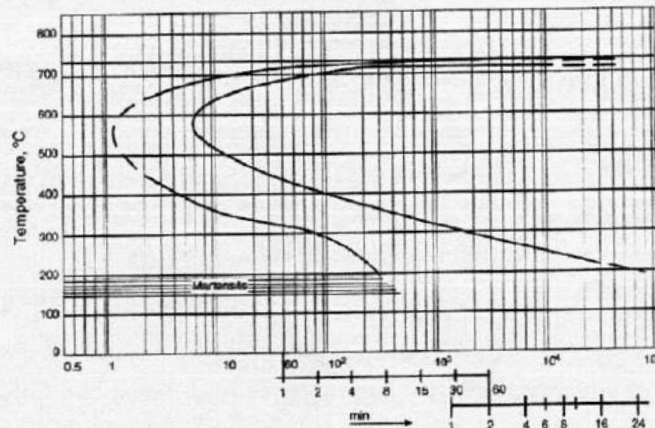


Figure 2(c)

3. (a) Briefly discuss the possible reasons of insufficient hardness of a heat treated product. (10)
- (b) Explain the process of Jominy End Quench hardenability test along with its advantages. (10)
- (c) Compare and construct the structures and properties obtained in ADIs with those obtained in (15)
quench and tempered ductile irons.
4. (a) Using Grossmann's method, demonstrate how you would determine the hardenability of plain (15)
carbon steel.
- (b) Describe the recrystallization mechanism of a cold-worked sample with necessary figure. (10)
- (c) Determine the carbon-equivalent of the following steels and comment on heating rate and (10)
sensitivity.

AISI-SAE	% C	% Mn	% Si	% Cr	% Mo
1095	0.9	0.3	-	-	-
4140	0.4	0.8	0.3	0.8	0.2

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Write down the advantages and disadvantages of Gas carburizing. (10)
- (b) Define Pilling Bedworth ratio and explain its effect on passivation action in the post oxidation (10)
of a Nitrided sample.

- (c) A mild steel shaft has 0.15% C, which needs to be carburized by pack carburizing method at 900°C . If the minimum carbon content at 2 mm from the surface is to be 0.3%, what will be carburizing time? Use figure 5(c) if necessary. (15)

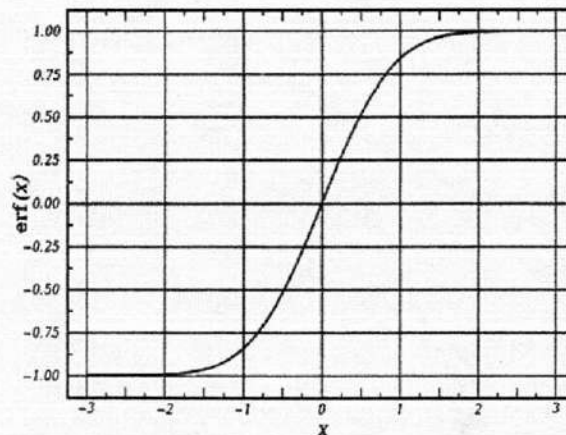


Figure 5(c)

6. (a) Explain how surface hardening advances over through hardening of a sample. Hence, list out different approaches of surface hardening mechanisms. (13)
- (b) Among grey, white, malleable and nodular cast irons, which one will you suggest for economic and mass production of high strength agricultural tools and why? (12)
- (c) A typical nodular cast iron component having 1.4% Silicon was austenitized at 780°C for hardening heat treatment. Find out the combined carbon content expected in the cast iron component after the prescribed heat treatment. (10)
7. (a) Compare and construct thermomechanically treated rebar with respect to conventional rebar from metallurgical perspective. (10)
- (b) Describe the heat treatment schedule of white cast irons. (10)
- (c) Mention the limitations that are quite unavoidable during the heat treatment of SS. Describe the heat treatment processes of martensitic SS along with their purposes. (15)
8. (a) How to identify if an alloy can be hardened by precipitation hardening mechanism? Explain with necessary diagram. (15)
- (b) The aging treatment of an Al-0.8Si-0.5Mg alloys are shown in the following Figure 8(b). Suggest an economically suitable aging treatment schedule for getting at least 350 MPa YS along with better age hardening flexibility. (10)

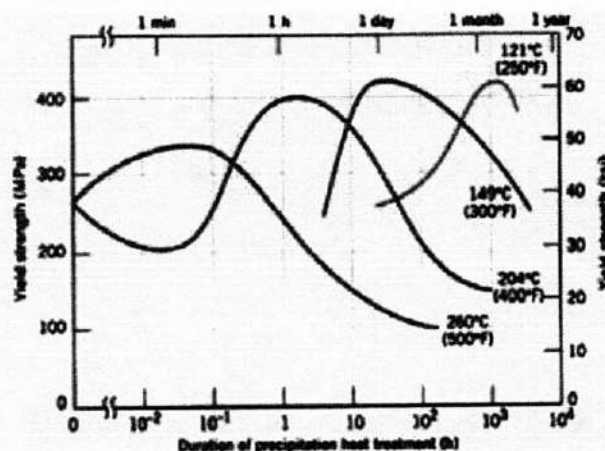


Figure 8(b)

- (c) Why superalloys possess FCC crystal structure? Describe the productions process of a TMT bar with appropriate diagram. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 1st Term Examination, 2022
Department of Materials Science and Engineering

Hum 3127
(Government and Sociology)

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) What is meant by politics? Is politics essential for the development of a country? Why? (15)
(b) Discuss the constituent elements of a state. Do the followings come under your definition of a state? (i) New York, (ii) UNO. (10)
Give reason for your answer.
(c) What is state? Is state necessary? Discuss. (10)
2. (a) Critically discuss the different types of government. What type of government do you like most? Why? (20)
(b) How to be a good citizen? "Some people have suggested that helping needy people is an important part of being a good citizen"-do you agree? Why? (15)
3. (a) Explain the origin and evolution of state according to the social contract theory. (10)
(b) What are the main features of Marxism? What were Marx's main contributions for understanding society and struggle of liberation? (15)
(c) Explain that "Fascism is not based on doctrine but reality". (10)
4. (a) What is constitution? What are the main characteristics of Bangladesh constitution? (10)
(b) Why do we need executive, legislature and judiciary? (15)
(c) What is socialism? What are the advantages and disadvantages of socialism? (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Define Sociology. What are the scopes of sociology? (10)
(b) Whose contribution is deemed to be the most influential for the development of sociology? (15)
(c) How did industrial revolution inspire for the development of sociology? Explain. (10)
6. (a) What do you understand by society and community? Which factors differentiate society from community? (10)
(b) "Society as a web of social relationship"-critically evaluate this idea of MacIver. (10)
(c) Why is social stratification universal? Explain it with examples. (15)
7. (a) What is culture? How is culture different from civilization? (10)
(b) What are the factors responsible for social change in Bangladesh? (10)
(c) What do you think about "Ethnocentrism"? Why is cultural relativism important? (15)
8. (a) What is meant by urbanization? Explain the pattern of urbanization in developing country. (15)
(b) Find out the major urban problems in third world countries. Discuss any two of them. (10)
(c) What are the grounds of white collar crime on society? (10)